

NBR VAT Assistant

Bilingual Retrieval-Augmented Generation System for Bangladesh VAT Law

PROJECT OVERVIEW

A Bengali/English legal question-answering system designed to help users retrieve and understand information from Bangladesh VAT law documents. The system grounds answers in the source documents and provides page-level citations so users can trace an answer back to the relevant legal text.

WHAT IT DOES

- Bengali and English VAT-law question answering
- Page-level source citations for traceability
- Hybrid semantic + lexical document retrieval
- Reciprocal Rank Fusion (RRF) for combining retrieval signals
- Cross-encoder reranking to improve the ordering of retrieved passages
- Bilingual evaluation set for systematic testing
- Handling of legacy PDF encoding problems in the source corpus

TECHNICAL ARCHITECTURE

User question → dense embedding retrieval + BM25 retrieval → Reciprocal Rank Fusion → cross-encoder reranking → relevant legal passages → LLM answer generation with citations.

TECH STACK

Python • bge-m3 • ChromaDB • BM25 • Cross-Encoder • OpenAI API

IMPLEMENTATION / CONTRIBUTION

Implemented the retrieval pipeline, document-processing workflow, bilingual evaluation setup, and retrieval improvements. The project focuses on making legal-document retrieval more reliable rather than relying only on an LLM's parametric knowledge.

PROJECT REPOSITORY

<https://github.com/ashuvo25/nbr-vat-assistant>

AGRONEST

AI-Powered Agricultural Decision System & Farmer Community Platform

PROJECT OVERVIEW

A full-stack platform combining agricultural community features with AI-assisted crop support. It is designed as a digital environment where farmers can interact with one another while also using computer vision and LLM-based capabilities for agricultural decision support.

WHAT IT DOES

- Social platform with posts, likes, comments, and friend connections
- Crop disease detection using computer vision
- AI-assisted analysis of detected agricultural problems
- LLM integration for recommendations and natural-language guidance
- Full-stack web architecture connecting frontend, backend, AI services, and database

TECHNICAL ARCHITECTURE

Farmer-facing web interface → Next.js frontend → FastAPI backend → AI inference/services → PostgreSQL persistence. Computer-vision outputs and application context can be combined with LLM-based analysis for user-facing recommendations.

TECH STACK

Next.js • FastAPI • YOLO • LLM • PostgreSQL • TensorFlow

IMPLEMENTATION / CONTRIBUTION

Worked on the integration of the web application with AI components, combining computer-vision-based disease detection with LLM-assisted agricultural analysis. The project demonstrates full-stack development together with practical AI integration.

PROJECT REPOSITORY

https://github.com/ashuvo25/AgroNest_LLM-ML

Agro-Scout

AI-Enabled Autonomous Agricultural Rover

PROJECT OVERVIEW

A robotics project aimed at automating field-level agricultural inspection and sample collection. The rover combines an embedded computing platform, onboard cameras, computer vision, and robotic control to perform agricultural tasks with reduced manual intervention.

WHAT IT DOES

- Dual operating modes: autonomous and remote control
- Onboard camera-based crop disease detection
- Automated identification/inspection of infected crop areas
- Automated soil sample collection from affected areas
- Raspberry Pi-based embedded computing and control

TECHNICAL ARCHITECTURE

Onboard camera → computer-vision processing → disease/target identification → rover navigation/control → soil-sampling mechanism. Remote-control mode provides manual operation when autonomous operation is not appropriate.

TECH STACK

Raspberry Pi • Computer Vision • IoT • Robotics • Python

IMPLEMENTATION / CONTRIBUTION

Contributed to the software/AI side of an agricultural robotics workflow, connecting computer vision with embedded robotic operation. The project demonstrates the application of AI beyond software-only systems by connecting perception to physical field operations.

PROJECT REPOSITORY

<https://github.com/Rifat-Rahman06/Agro-Scout>

UIU Assistant - RAG Chatbot

Retrieval-Augmented Chatbot for United International University

PROJECT OVERVIEW

An intelligent university-information chatbot that uses current content from the United International University website as its knowledge source. Instead of depending only on an LLM's pretrained knowledge, the system crawls university webpages, indexes the content, retrieves relevant information, and uses an LLM to formulate the response.

WHAT IT DOES

- Automated web crawling and content collection
- Website content indexing for retrieval
- Semantic search using a vector database
- Retrieval-Augmented Generation architecture
- Groq-powered LLM response generation
- Designed around live/current university website information

TECHNICAL ARCHITECTURE

UIU website → web crawler → document processing/indexing → Pinecone vector database → semantic retrieval → relevant context → Groq LLM → natural-language answer.

TECH STACK

Python • LangChain • Pinecone • Groq • Web Scraping • RAG

IMPLEMENTATION / CONTRIBUTION

Built the core RAG workflow around crawled university information, including content retrieval, vector-based semantic search, and LLM response generation. The project demonstrates how external, changing web knowledge can be connected to an LLM through retrieval.

PROJECT REPOSITORY

<https://github.com/Rifat-Rahman06/UIU-assistant>